

Chapter 3

Random Variables and Distributions

Learning Objectives

- Discuss random variables
- Determine probabilities from probability density functions.
- Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse.

3.0 Random Variables and Distribution

3.1 Introduction

Experiments are conducted with results that are subject to chance. Suppose that to each point of a sample space we assign a number, we then have a function defined on the sample space. This function is called a random variable (or stochastic variable). In other words, in a particular experiment a random variable X would be some function that assigns a real number $X(s)$ to each possible outcome $s \in S$.

Definition: Let S be the sample space associated with some experiment ε . A random variable X is a function that assigns a real number $X(s)$ to each sample point $s \in S$.

There are two basic types of random variables: the Discrete random variable and the Continuous random variable.

3.2 Discrete Random Variable

A **discrete** random variable is a random variable whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on (“countably” infinite).

3.1.1 Probability Distribution of Discrete Random Variable

For a random variable X of the discrete type, it is necessary and sufficient to know the following:

- (a) the list of all possible values x_i , $i = 1, 2, 3, \dots$
- (b) the corresponding probability $p(X = x_i)$ of each of these values.

When such information is available we say that the distribution law or the probability distribution or simply the distribution of the random variable X is known.

Definition: A probability distribution of a discrete random variable X is a sequence of the values of X , together with the probability assigned to each point x_i , ($i = 1, 2, 3, \dots$) x_i , $i = 1, 2, 3, \dots$

The probability distribution of a discrete random variable is more often called probability function ($p.f.$) or a probability mass function ($p.m.f.$) and is frequently denoted by $p(x_i)$ and $P(X = x_i)$.

Representation of Probability Distribution of the Discrete Random Variable

The probability distribution of a discrete random variable X can be represented by a table, a graph or a function.

Tabular form

It is convenient to specify the probability distribution of a random variable by means of a table having two rows: the upper row contains the possible values the random variable assumes and the lower row contains the corresponding probabilities of the values. This is represented as in the table 1 below:

Table 1: Probability distribution of X

x_i	x_1	x_2	$\dots$	x_n
$p(x_i)$	$p(x_1)$	$p(x_2)$	$\dots$	$p(x_n)$

Graphical form

The probability distribution may also be given graphically. The graph represents chance and not data.

Definition: A graph of $p(x_i)$ against x_i is called a probability graph.

To obtain a probability graph, vertical lines or bars are drawn above the possible values x_i of the random variable X on the horizontal axis. The height of each line or bar (rectangle) equals the probabilities of the corresponding values of the random variables.

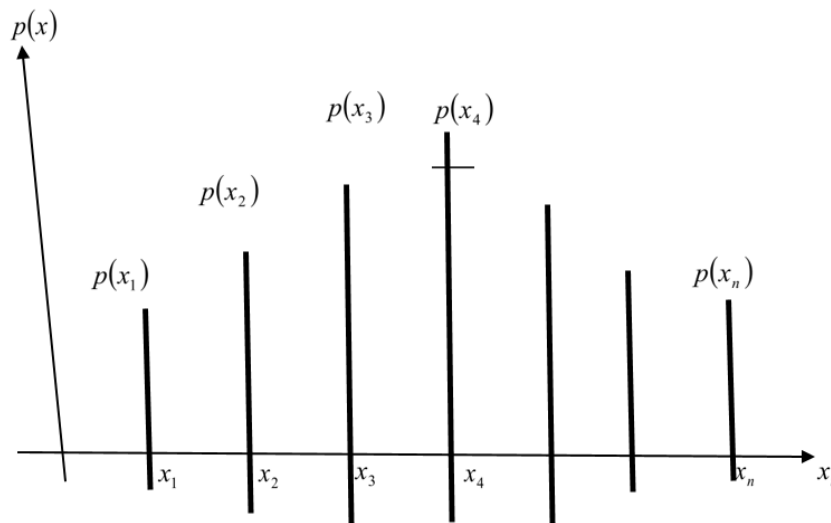


Figure 3.1: Figure 1: A graph of a probability distribution

Formula:

Representing a discrete probability by means of a formula is the most useful method.

For example:

$$p(x) = P(X = x) = \frac{1}{n}, \quad x = 1, 2, 3, \dots, n$$

defines a probability distribution of a random variable X . Not all functions of a discrete random variable qualify to be called probability mass function. We shall now give a formal definition of a probability mass function.

Definition: Probability mass function: Any function $p(x)$ defined on all possible values of the discrete random variable $X = x_i$, $i = 1, 2, 3, \dots$ is called a probability mass function if it satisfies the following properties:

$$\begin{aligned} \text{property 1} \quad & p(x_i) \geq 0, \quad i = 1, 2, 3, \dots \\ \text{property 2} \quad & \sum_i p(x_i) = 1 \end{aligned}$$

where the summation is over all possible values of the random variable X .

A random variable X in one experiment takes only one of its n possible values with a corresponding probability which is non-negative. Thus, in an experiment, one and only one of the numerous possible mutually exclusive events will happen. Consequently, the sum of the probabilities of these events:

$$p(x_1) + p(x_2) + p(x_3) + p(x_4) + \dots + p(x_n) = 1$$

Example 25

A fair coin is tossed three times. Let the random variable X represent the number of heads which come up.

- find the probability distribution corresponding to the random variable X
- Construct a probability graph.

Solution

- the sample space is

$$S = \{HHH, THH, HTH, HHT, HTT, THT, TTH, TTT\}$$

The probability of each outcome is $\frac{1}{8}$, since all the outcomes are equally likely sample events. With each sample point we can associate a number for the random variable X , as shown in the table below:

Sample point	HHH	THH	HTH	HHT	HTT	THT	TTH	TTT
Number of Head	3	2	2	2	1	1	1	0
$p(X = x)$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$

The table above shows that the random variable can take the values 0, 1, 2, 3. The next task is to compute the probability distribution $p(x_i)$ of X . Thus,

$$p(3) = P(X = 3) = P(HHH) = \frac{1}{8}$$

$$\begin{aligned} p(2) &= P(X = 2) = P(\{HHT\} \cup \{HTH\} \cup \{THH\}) \\ &= P(\{HHT\}) + P(\{HTH\}) + P(\{THH\}) \\ &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8} \end{aligned}$$

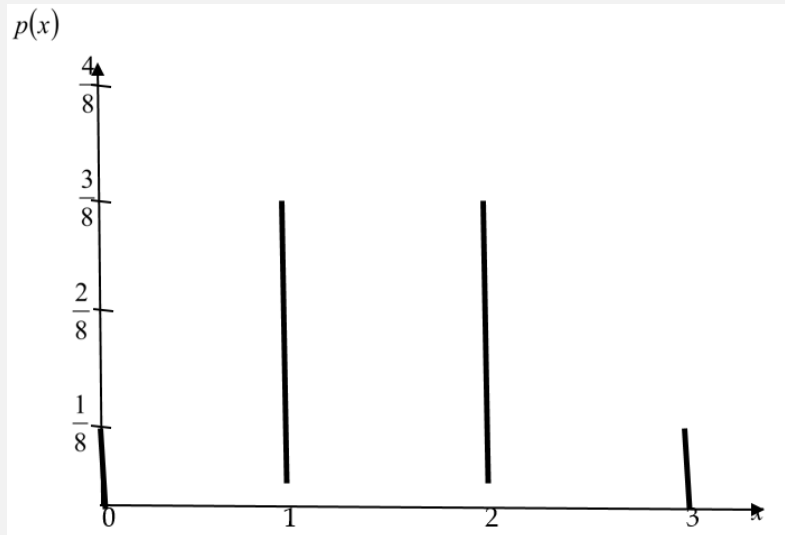
$$\begin{aligned}
 p(1) &= P(X = 1) = P(\{TTH\} \cup \{THT\} \cup \{HTT\}) \\
 &= P(\{TTH\}) + P(\{THT\}) + P(\{HTT\}) \\
 &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}
 \end{aligned}$$

$$p(0) = P(X = 0) = P(TTT) = \frac{1}{8}$$

Thus, the probability distribution of X is tabulated as:

x_i	0	1	2	3
$p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(b). The diagram which follows graphically describes the above distribution.



Example 26

A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes random purchase of 2 of these computers, find the probability distribution for the number of defectives

Solution

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can be any of the numbers 0, 1 and 2. Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0}\binom{5}{2}}{\binom{8}{2}} = \frac{10}{28}$$

$$f(1) = P(X = 1) = \frac{\binom{3}{1}\binom{5}{1}}{\binom{8}{2}} = \frac{15}{28}$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{5}{0}}{\binom{8}{2}} = \frac{3}{28}$$

Thus, the probability distribution of X is:

x	0	1	2
$f(x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$

As a check, we can see whether the three probabilities we found will sum up to one because as usual, the total probability associated with a number of mutually exclusive, exhaustive events must be one.

Note: A probability distribution is a display of all possible outcomes of an experiment along with the probabilities of each outcome. In fact, it is a list of all possible outcomes of some experiment and the probability associated with each outcome.

Example 27

Verify that the following probability distribution functions are probability mass functions.

(i).

$$p(x) = \begin{cases} \frac{1}{21}(2x + 3), & x = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

(ii).

$$p(x) = \begin{cases} k(x - 1), & x = 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Solution

(i). For the probability distribution function to be a probability mass function the following must satisfy:

$p(x) > 0$ for all x and

$$\begin{aligned} \sum_{x=1}^3 p(x) &= \frac{1}{21} \sum_{x=1}^3 (2x + 3) \\ &= \frac{1}{21} [\{2(1) + 3\} + \{2(2) + 3\} + \{2(3) + 3\}] \\ &= \frac{1}{21} [5 + 7 + 9] \\ &= 1 \end{aligned}$$

(ii). The value of k is determined by assuming that $p(x)$ is a probability function. Thus,

$$\begin{aligned} \sum_{x=3}^5 p(x) &= \sum_{x=3}^5 k(x - 1) = 1 \\ \Rightarrow k[(3 - 1) + (4 - 1) + (5 - 1)] &= 1 \\ \Rightarrow k[2 + 3 + 4] &= 1 \\ \Rightarrow 9k &= 1 \\ \Rightarrow k &= \frac{1}{9} \end{aligned}$$

3.3 Continuous Random Variable

If X is a continuous random variable, the probability that X takes on any one particular value is generally zero. Therefore, we cannot define a continuous random variable in the same way as for a discrete random variable. In order to arrive at a probability distribution for a continuous random variable we note that the probability that X lies between two different values is meaningful. Thus, a continuous random variable is the type whose spaces are not composed of a countable number of points but takes on values in some interval or a union of intervals of the real line.

Definition: If the set of all possible values of a random variable X , takes on an uncountably infinite number of values or values in some interval or a union of intervals of the real line, it is called a continuous random variable if there exists a function f , called probability density function of X such that

(i) $f(x_i) \geq 0 \forall x_i$ (Non-negative)

(ii)

$$\int_{-\infty}^{\infty} f(x_i) dx = 1$$

(iii)

$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad -\infty \leq a \leq b \leq \infty$$

3.2.1 Probability Distribution of Continuous Random Variable

Continuous random variable can be considered as one that takes on values in some interval of the real line. The probability distribution for a continuous random variable is more often called a probability density function (p.d.f) or simply density function and is denoted by $f(x)$.

Definition: The probability density function, (p.d.f.) of a random variable X of the continuous type, with space R that is an interval or union of intervals, defined on the set of real numbers is an integrable function $f(x)$ satisfying the following conditions.

(i) $f(x_i) \geq 0 \forall x_i$ (Non-negative)

(ii)

$$\int_{-\infty}^{\infty} f(x_i) dx = 1$$

(iii) The probability of the event $X \in A$ is

$$P(X \in A) = \int_A f(x) dx$$

A typical probability function of the continuous type is as shown on the sketch below

$$f(x)$$

0 a b X

fig. 2.1 An example of a Probability Density Function

In this figure, the total area under the curve must be equal to 1 and the value of the $P(a \leq x \leq b)$ is equal to the area of the shaded region.

Example 28

Suppose that the error in the reaction temperature, in $^{\circ}C$ for a controlled laboratory experiment

is a continuous random variable X having the probability density function:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Verify (a). $\int_{-\infty}^{\infty} f(x) dx$ (b). $P(0 < x \leq 1)$

Solution

(a).

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-1}^2 \frac{x^2}{3} dx \\ &= \frac{x^3}{9} \Big|_{-1}^2 = \frac{8}{9} + \frac{1}{9} = 1 \end{aligned}$$

(b).

$$\begin{aligned} P(0 < x \leq 1) &= \int_0^1 \frac{x^2}{3} dx \\ &= \frac{x^3}{9} \Big|_0^1 = \frac{1}{9} - 0 = \frac{1}{9} \end{aligned}$$

Example 29

(a). Find the constant C such that the function below is a probability density function:

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(b). Compute $P(1 < x < 2)$

Solution

(a).

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^3 cx^2 dx \\ &= \frac{cx^3}{3} \Big|_0^3 = 9c \end{aligned}$$

But since

$$\int_{-\infty}^{\infty} f(x) dx = 1, \quad 9c = 1 \quad \therefore c = \frac{1}{9}$$

(b).

$$\begin{aligned} P(1 < x \leq 2) &= \int_1^2 \frac{x^2}{9} dx \\ &= \frac{x^3}{27} \Big|_1^2 = \frac{8}{27} - \frac{1}{27} = \frac{7}{27} \end{aligned}$$

Example 30

A machine produced copper wire, and occasionally there is a flaw at some point along the wire. The length of wire (in meters) produced between successive flaws is a continuous random variable X with p.d.f of the form

$$f(x) = \begin{cases} c(1+x)^{-3} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

where c is a constant

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^{\infty} c(1+x)^{-3} dx = 1$$

Let $U = (1 + x)$ and $du = dx$ apply power rule for integral

$$c \int_0^{\infty} U^{-3} du = 1$$

$$c \left[\frac{U^{-2}}{-2} \right]_0^{\infty} = 1$$

$$c \left[\frac{1}{2} \right] = 1$$

$$c = 2$$

Example 31

For each of the following functions, find the constant c so that $f(x)$ is a p.d.f of a random variable X .

$$(i) f(x) = 4xc, 0 \leq x \leq 1$$

$$(ii) f(x) = c\sqrt{x}, 0 \leq x \leq 4$$

$$(iii) f(x) = c/x^{3/4}, 0 < x < 1$$

Solution

$$(i) f(x) = 4x^c$$

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 4x^c dx = 1$$

$$4 \int_0^1 x^c dx = 1$$

$$4 \left[\frac{x^{c+1}}{c+1} \right]_0^1 = 1$$

$$4 \left[\frac{1}{c+1} \right] = 1$$

$$c+1 = 4$$

$$c = 3$$

$$(ii) f(x) = c\sqrt{x}$$

(iii)

$$c \int_0^4 x^{1/2} dx = 1$$

$$c \left[\frac{x^{1/2+1}}{1/2+1} \right]_0^4 = 1$$

$$c \left[\frac{4^{3/2}}{3/2} \right] = 1$$

$$c = 3/16$$

$$f(x) = \frac{c}{x^{3/4}}$$

$$c \int_0^1 x^{-3/4} dx = 1$$

$$c \left[\frac{x^{-3/4+1}}{-3/4+1} \right]_0^1 = 1$$

$$\frac{c}{1/4} = 1$$

$$c = \frac{1}{4}$$

Post-Test

1. Define the following terms;
2. discrete random variable
3. continuous random variable
4. Let $f(x) = \frac{x}{15}$, $x = 1, 2, 3, 4, 5$, zero elsewhere be the p.d.f. of X. Find;
5. $\Pr [1 \text{ or } 2]$

$$\Pr[1 < X < 3]$$

$$(i) \Pr \left[\frac{1}{2} \leq X \leq \frac{5}{2} \right]$$

- (ii) For each of the following functions, find the constant c so that $f(x)$, satisfies the conditions of being a p.d.f. of a random variable X.

$$(i) f(x) = cx^3, 0 < x < 1$$

(ii)

$$f(x) = \frac{c}{x^4}$$

,

$$0 < x < \infty$$

3.4 Cumulative Distribution Function

The cumulative distribution function (c.d.f.) or simply the distribution function is the most universal characteristic of a random variable. It exists for all random variables whether they are

discrete or continuous.

Definition: Let X be a random variable and x any real number. The cumulative distribution function of X is a function $F(x)$ defined as the probability that the random variable X takes a value less than or equal to x such that

$$F(x) = P(X \leq x)$$

The function $F(x)$ in the above definition may also be written as

$$F(x) = P(-\infty < X \leq x)$$

3.3.1 Cumulative distribution function of Discrete Random Variables

The statement of the theorem below for a distribution function of a discrete random variable follows trivially from the definition of cumulative distribution function.

Theorem (Distribution function of a discrete random variable): Let X be a discrete random variable with probability $p(x_i)$, then the cumulative distribution function is defined by

$$F(x) = \sum_{x_i \leq x} p(x_i)$$

For the discrete case, the random variable X takes on only a finite number of values $x_1, x_2, x_3, \dots, x_n$. In this case, the distribution function $F(x)$ is given by

$$F(x) = \begin{cases} 0 & -\infty \leq x < x_1 \\ p(x_1) & x_1 \leq x < x_2 \\ p(x_1) + p(x_2) & x_2 \leq x < x_3 \\ \vdots & \vdots \\ p(x_1) + p(x_2) + \dots + p(x_n) & x_n \leq x < +\infty \end{cases}$$

The figure below depicts the graph $F(x)$ which is discontinuous at at the possible values $x_1, x_2, x_3, \dots, x_n$. At these points $F(x)$ is continuous from the right but discontinuous from the left. Because of the appearance of its graph, the cumulative distribution function for a discrete random variable is also called a staircase function or a step function, having jump discontinuities at all possible values $X = x_i$ with a step at the x_i of height $p(x_i)$. The graph increases only through these jumps at $x_1, x_2, x_3, \dots, x_n$. Everywhere else in the interval $[x_i, x_{i+1}]$, the cumulative distribution function $F(x_i)$ is constant.

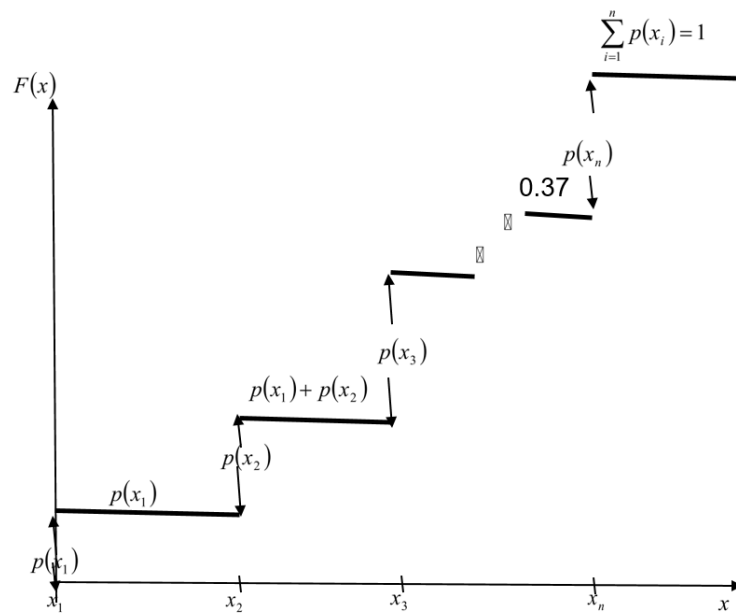


Figure 3.2: Figure 2 Cumulative Distribution Function (Discrete Case)

Example 32

A fair coin is tossed three times. Let X represent the number of heads which come up

- find the cumulative distribution function and
- Sketch the graph.

Solution

- To obtain the cumulative distribution function, we need the following steps:

Step 1:

Find the probability distribution of the random variable X . The probability distribution of this example has been found in an earlier example and produces the results here for convenience.

x_i	0	1	2	3
$p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Step 2:

Find the cumulative distribution function:

$$\begin{aligned}
 F(0) &= p(X \leq 0) = p(0 \leq X < 1) \\
 &= p(X < 0) + p(X = 0) \\
 &= 0 + p(0) \\
 &= \frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(1) &= p(X \leq 1) = p(0 \leq X < 2) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) \\
 &= 0 + p(0) + p(1) \\
 &= 0 + \frac{1}{8} + \frac{3}{8} \\
 &= \frac{4}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(2) &= p(X \leq 2) = p(0 \leq X < 3) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) + p(X = 2) \\
 &= 0 + p(0) + p(1) + p(2) \\
 &= 0 + \frac{1}{8} + \frac{3}{8} + \frac{3}{8} \\
 &= \frac{7}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(3) &= p(X \leq 3) = p(0 \leq X \leq 3) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) + p(X = 2) + p(X = 3) \\
 &= 0 + p(0) + p(1) + p(2) + p(3)
 \end{aligned}$$

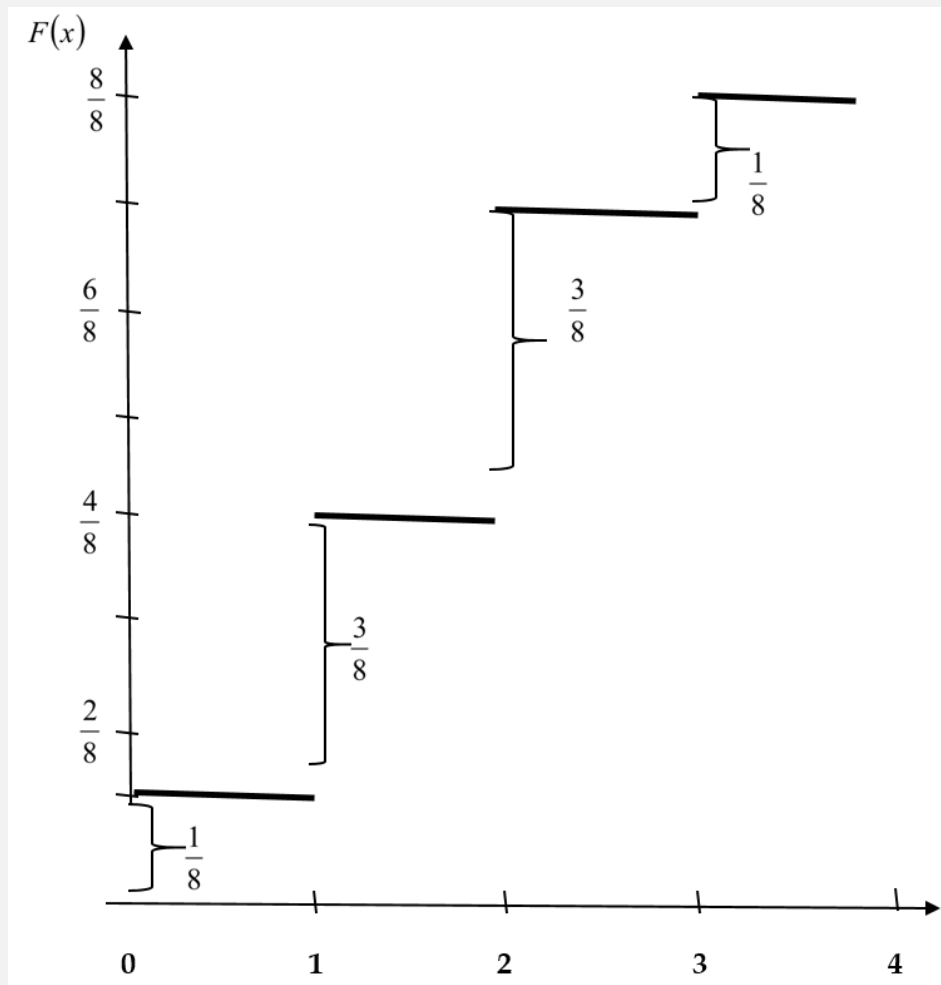
$$= 0 + \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8}$$

$$= 1$$

Hence the cumulative distribution function is

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{4}{8} & 1 \leq x < 2 \\ \frac{7}{8} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

(b), The graph of $F(x)$ is shown in the figure below:



Finding the probability distribution from the cumulative distribution function is a straightforward situation. If $F(x)$ is the cumulative distribution function of a discrete random variable X , we then find the points at which the cumulative distribution function jumps, and the jump sizes. The probability function has masses exactly at those jump points, with the probability masses being equal in magnitude to the respective jump sizes. It is for this reason that it is called probability mass function.

Example 33

Suppose you are given the cumulative distribution function below:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{4}{8} & 1 \leq x < 2 \\ \frac{7}{8} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Find its probability distribution.

Solution

It would be noted from the graph of the cumulative distribution function that the magnitudes or the heights (that is, $p(x_i)$) of the jumps (steps) at 0, 1, 2, 3 are $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ respectively, hence

$$p(x) = \begin{cases} \frac{1}{8} & x = 0 \\ \frac{4}{8} & x = 1 \\ \frac{3}{8} & x = 2 \\ \frac{1}{8} & x = 3 \end{cases}$$

Note:

We can obtain this result without the graph by finding the difference in the adjacent values of $F(x)$.

3.3.2 Cumulative distribution function of Continuous Random Variables

Again, the statement of the theorem below for a distribution function of a continuous random variable follows trivially from the definition of cumulative distribution function.

Theorem (Distribution function of a continuous random variable): Let X be a continuous random variable with probability distribution function $f(x)$, Then the cumulative distribution function $F(x)$ is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \quad \text{for } -\infty < x < \infty$$

The typical graph of the cumulative distribution function of the continuous random variable is as shown below.

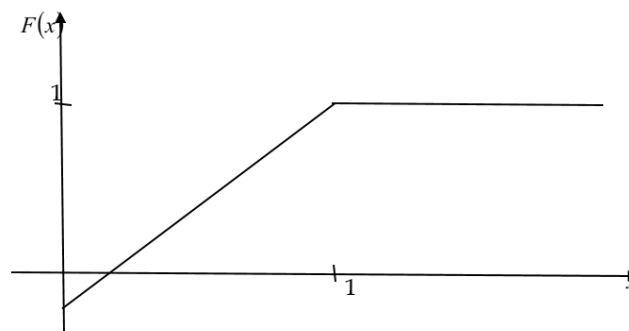


Figure 3.3: Figure 2 Cumulative Distribution Function (Continuous Case)

The graph of $F(x)$ is continuous. Its slope need not be everywhere continuous, but where this is so, it is equal to the probability density function. Thus, $f(x) = \frac{d}{dx}F(x)$ if the derivative exists and $P(a < X < b) = F(b) - F(a)$.

Example 34

The probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x \leq 2 \\ 0 & x > 2 \end{cases}$$

Find the cumulative distribution function and sketch its graph.

Solution

If $x < 0$, then

$$F(x) = \int_{-\infty}^x f(t) dt = 0$$

If $0 < x < 2$, then

$$\begin{aligned} F(x) &= \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt \\ &= 0 + \int_0^x \frac{t}{2} dt = 0 + \int_0^x \frac{t}{2} dt \\ &= 0 + \left. \frac{t^2}{4} \right|_0^x = \frac{x^2}{4} \end{aligned}$$

If $x > 2$, then

$$\begin{aligned} F(x) &= \int_{-\infty}^0 f(t) dt + \int_0^2 f(t) dt + \int_2^x f(t) dt \\ &= 0 + \int_0^2 \frac{t}{2} dt + \int_2^x f(t) dt \\ &= 0 + \int_0^2 \frac{t}{2} dt + \int_2^x f(t) dt \\ &= 0 + \left. \frac{t^2}{4} \right|_0^2 + 0 = 1 \end{aligned}$$

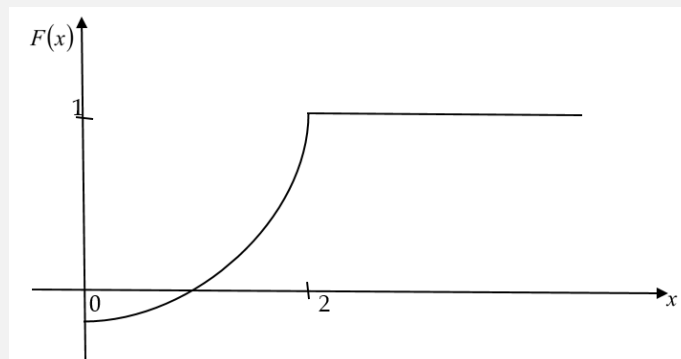


Figure 2 Cumulative Distribution Function (Continuous Case)